

Specification of a Multi-Agent System for Parkinson's Disease Analysis

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1 Introduction

[TODO]

2 Biological Background

Parkinson's disease is the second most common neurodegenerative disorder following Alzheimer's disease. The disease is classically characterized by overt motor symptoms which include slowness of movement, muscle rigidity and resting tremors. Furthermore, patients often experience debilitating non-motor symptoms that may manifest prior to the onset of motor problems.

2.1 Substantia Nigra

While Parkinson's disease affects various regions of the nervous system and multiple types of neurons, a primary focal point is the *substantia nigra pars compacta*, located in the midbrain. This region is a crucial component of a major neural pathway responsible for facilitating movement. The progressive death of dopaminergic neurons within the substantia nigra leads to the malfunction of this pathway, directly resulting in the deficits observed in patients.

2.2 Protein Misfolding

A distinctive pathological mark found in the majority of Parkinson's cases is the presence of intracellular clumps of misfolded proteins, with the most common type being Lewy bodies. A key structural component of these aggregates is a misfolded protein known as α -synuclein. These misfolded proteins are capable of forming small repeated units called oligomers or developing into longer fibrous structures. Scientific evidences suggest that these forms of α -synuclein are highly toxic to neurons, overwhelming the cellular system responsible for protein-degradation and leading to neuronal death.

2.3 Mitochondrial dysfunction

Mitochondria are the organelles responsible for providing cells with the energy necessary for vital functions, they are also deeply implicated in the pathogenesis of Parkinson's disease, as they are responsible for critical processes that become impaired in Parkinson's disease. Furthermore, the vital recycling process designed to remove and replace old or damaged mitochondria is thought to be severely disrupted, resulting in toxic accumulation of dysfunctional mitochondria that can no longer sustain neuronal health.

2.4 Neuroinflammation and Glial Cells

The role of glial cells is another crucial aspect of Parkinson's pathology. As dopaminergic neurons die off, specialized glial cells known as microglia are believed to clear the resulting cellular debris. This action triggers an immune response. Once activated, microglia release inflammatory cytokines that activate neighboring microglia, as well as astrocytes. The environment created by these activated cells promotes further neuroinflammation and cellular damage.

2.5 Disease spread and future therapeutic targets

As the disease progresses over time, pathology develops across more areas of the nervous system. An emerging hypothesis is that this widespread progression occurs through the direct transmission of misfolded α -synuclein between cells. Current active areas of research include developing methods to enhance the clearance of abnormal proteins, blocking cell-to-cell transmission of misfolded proteins, improving mitochondrial function, and targeting the underlying neuroinflammation.

3 Intra-cellular Environment

The **Intra-cellular environment** is the environment within a singular dopaminergic neuron. It represents the general neuron health and the way the organelles degrade it.

It is defined as a triple $Env_{intra} = \langle E, e_0, \tau \rangle$, where E is a set of states (represented as vector) in the format of:

$$\begin{aligned}
 e \in E : & \langle V_{energy}, V_{toxicity}, V_{clearance}, V_{stress}, V_{concentration}, V_{workload}, V_{target} \rangle \\
 & V_{energy} \in \{0: \text{Low}, 1: \text{High}\} \\
 & V_{toxicity} \in \{0: \text{Clear}, 1: \text{Toxic}\} \\
 & V_{clearance} \in \{0: \text{Available}, 1: \text{Overwhelmed}\} \\
 & V_{stress} \in \{0: \text{Normal}, 1: \text{High}\} \\
 & V_{concentration} \in \{0: \text{Low}, 1: \text{High}\} \\
 & V_{workload} \in \{0: \text{Normal}, 1: \text{High}\} \\
 & V_{target} \in \{0: \text{NonToxic}, 1: \text{Toxic}\}
 \end{aligned}$$

e_0 is the initial state of the environment, defined as $\langle 0, 0, 0, 0, 0, 0, 0 \rangle$ (**Healthy State**).

3.1 Intra-cellular Agents

All the agents are adaptive agents, which are defined by a 9-tuple $\langle I, E, Per, Act, i_0, see, next, do, action \rangle$.

- I is the set of internal states for the agent;
- E is the set of possible environmental states;
- Per is a partition of E that represents what the agent perceives
- Act is a set of possible actions, i_0 is the initial state of the agent;
- $see : E \rightarrow Per$ allows the agent to observe the environment;
- $action : I \times Per \rightarrow Act$ identifies the correct action;
- $do : Act \times E \rightarrow E$ applies the action chosen to the environment;
- $next : I \times Per \rightarrow I$ transition into the new state;

Given this description, multiple agents were defined.

3.1.1 Ag_{mito} (Mitochondria)

Internal states

$I : \{0 : \text{Healthy}, 1 : \text{Consumed}, 2 : \text{Damaged}, 3 : \text{Debris Overwhelmed}\}$

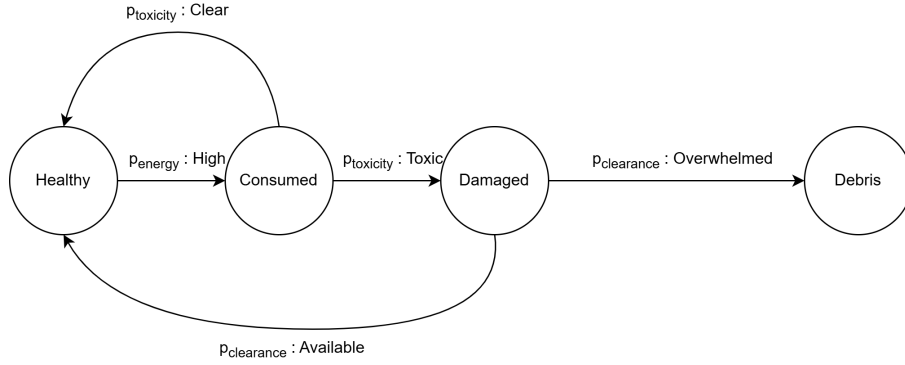
Possible actions

$Act : \{Fusion, Fission, Transport, Produce_ATP, Recycle, Fail_and_impair\}$

Perception

$Per = \langle p_{energy} : \{Low, High\}, p_{toxicity} : \{Clear, Toxic\}, p_{clearance} : \{Available, Overwhelmed\} \rangle$

Next automaton



Action mapping

$$\begin{aligned}
 action(Healthy, (High, -, -)) &= Produce_{ATP} \wedge Fission \\
 action(Consumed, (High, Clear, -)) &= Fusion \\
 action(Damaged, (-, -, Available)) &= Recycle \\
 action(Damaged, (-, -, Overwhelmed)) &= Fail_and_impair
 \end{aligned}$$

3.1.2 Ag_α (α -synuclein)

Internal states

$I : \{0 : Monomer, 1 : Misfolded, 2 : Oligomer, 3 : LewyBodies, 4 : Cleared\}$

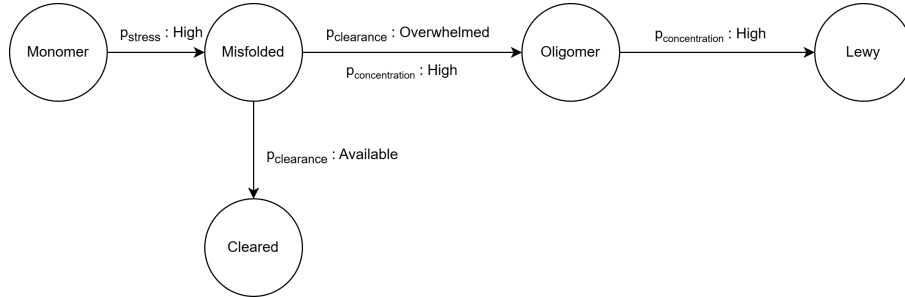
Possible actions

$Act : \{Misfold, Degrade, Aggregate, Intoxicate, Transmit\}$

Perception

$Per = \langle p_{stress} : \{0 : Normal, 1 : High\}, p_{clearance} : \{0 : Available, 1 : Overwhelmed\}, p_{concentration} : \{0 : Low, 1 : High\} \rangle$

Next automaton



Action mapping

$$\begin{aligned}
 action(Misfolded, (High, -, -)) &= Misfold \\
 action(Misfolded, (-, Available, -)) &= Degrade \\
 action(Misfolded, (-, Overwhelmed, -)) &= Aggregate \\
 action(Oligomer, (-, -, -)) &= Intoxicate \wedge Aggregate
 \end{aligned}$$

3.1.3 Ag_{deg} (Degrading Machinery)

Internal states

$I : \{0 : Inactive, 1 : Active, 2 : Stressed, 3 : Overwhelmed\}$

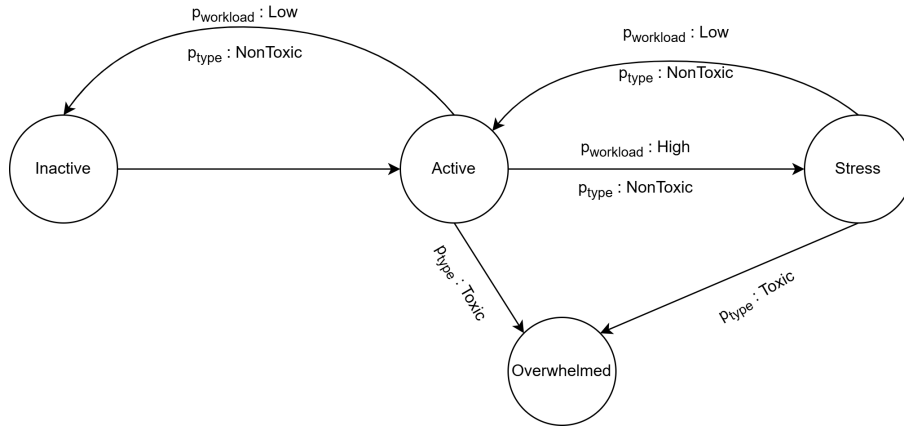
Possible actions

$Act : \{Scan, Degrade, Upregulate, Fail\}$

Perception

$Per : \langle p_{workload} : \{Low, High\}, p_{target} : \{NonToxic, Toxic\} \rangle$

Next automaton



Action mapping

$action(Inactive, (-, -)) = Scan$

$action(Active, (-, -)) = Degrade$

$action(Stress, (-, -)) = Upregulate \wedge Degrade$

$action(Overwhelmed, (-, -)) = Fail$

4 Extra-cellular environment

The **Extra-cellular environment** is the environment outside the singular dopaminergic neuron, where multiple neurons interact with each others and with their neighbours.

Likewise the intra-cellular environment, it is defined as a triple $Env_{extra} = \langle E, e_0, \tau \rangle$, where E is a set of states (represented as vectors) in the format of:

$$\begin{aligned} e \in E : & \langle V_{debris}, V_{cytokines}, V_{neurotoxins}, V_{support}, V_{dopamine} \rangle \\ V_{debris} & \in \{Low, High\} \\ V_{cytokines} & \in \{Basal, Storm\} \\ V_{neurotoxins} & \in \{Clear, Toxic\} \\ V_{support} & \in \{Provided, Withdrawn\} \\ V_{dopamine} & \in \{Optimal, Reduced, Critical\} \end{aligned}$$

4.1 Ag_{mic} (Microglia Agent)

Internal states

$$I : \{0 : Resting, 1 : Phagocytic, 2 : Activated_M1\}$$

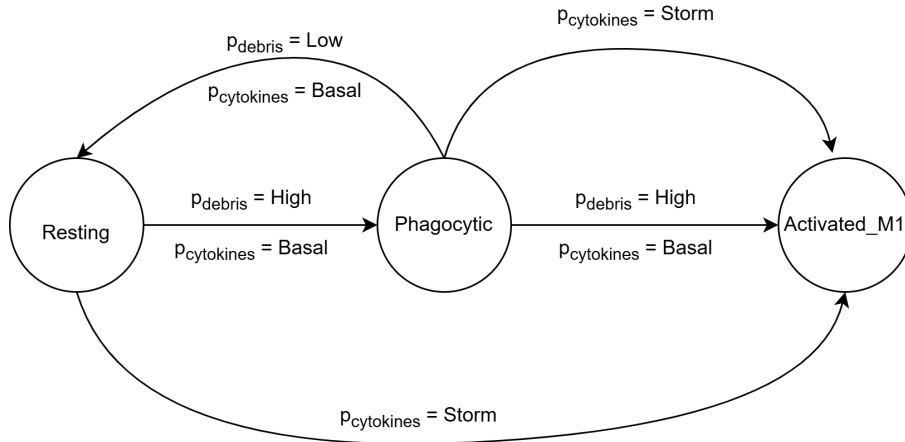
Possible actions

$$Act : \{Surveil, Clear_Debris, Release_Cytokines\}$$

Perception

$$Per : \{p_{debris} : \{Low, High\}, p_{cytokines} : \{Basal, Storm\}\}$$

Next automaton



Action mapping

$$\begin{aligned} action(Resting, (-, -)) &= Surveilance \\ action(Phagocytic, (-, -)) &= Clear_Debris \\ action(Activated_M1, (-, -)) &= Release_Cytokines \end{aligned}$$

4.2 Ag_{astro} (Astrocyte Agent)

Internal states

$$I : \{0 : Supportive, 1 : Reactive\}$$

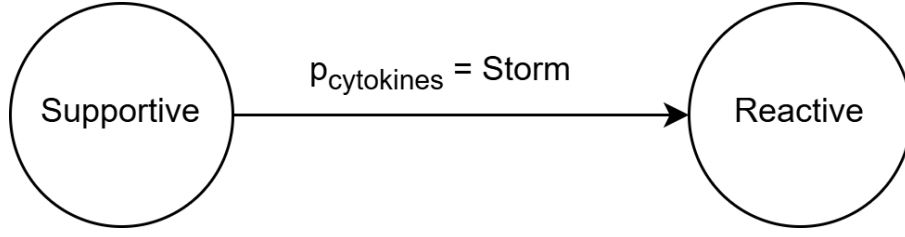
Possible actions

$$Act : \{Provide_Support, Withdraw_Support, Release_Neurotoxins\}$$

Perception

$$Per : \{p_{cytokines} : \{Basal, Storm\}\}$$

Next function



Action mapping

$action(Supportive, -) = Provide_Support$

$action(Reactive, -) = Withdraw_Support \wedge Release_Neurotoxins$

4.3 Ag_{neuron} (Dopaminergic Neuron Agent)

It acts as a hierarchical automaton, it perceives the macroscopic environment, but its survival depends on its internal environment (Env_{intra}). When the internal environment reaches the *Dead* state, the neuron undergoes apoptosis, ruptures and spills its toxic intracellular contents into the extracellular space.

Internal states

$I : \{Healthy, Compromised, Apoptotic, Ruptured\}$

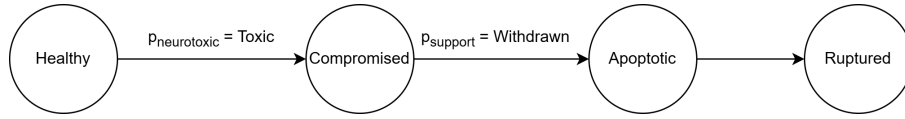
Possible actions

$Act = \{Release_Dopamine, Signal_Failure, Absorb_Toxins, Dump_Debris\}$

Perception

$Per = \{p_{neurotoxins} : \{Clear, Toxic\}, p_{support} : \{Provided, Withdrawn\}\}$

Next automaton



Action mapping

$action(Healthy, (-, -, -)) = Release_Dopamine$

$action(Compromised, (-, -, -)) = Signal_Failure \wedge Absorb_Toxins$

$action(Apoptotic, (-, -, -)) = Signal_Failure$

$action(Ruptured, (-, -, -)) = Dump_Debris$